power

File: power.kicad_sch

pi_link

File: pi_link.kicad_sch

radio

File: radio.kicad_sch

ui

File: ui.kicad_sch

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DRAFT - for ET review
MTP F26 Team C (UPC)

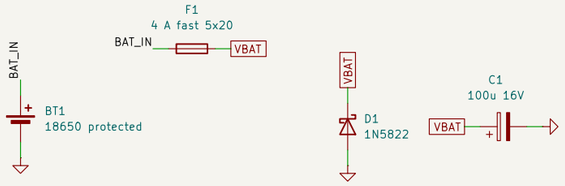
Sheet: /
File: teamc-carrier-simple.kicad_sch

Title: Team C carrier rev C - simple, UPC lab, through-hole + modules (DRA

Size: A4	Date: 2026-10-24	**Rev: DRAFT**
KiCad E.D.A. 10.0.6		Id: 1/5

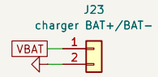
POWER - one protected 18650 cell → 5 V for the Pi (Pololu module) and 3.3 V for the radio (MCP1700)

1 CELL, FUSE AND REVERSED-CELL PROTECTION



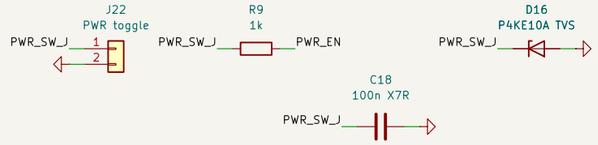
What it does, in plain words:
BT1 – the cell sits in a holder soldered on this board (no battery lead). Use a PROTECTED 18650 (its own protection circuit cuts at a short or a deep discharge). A cable tie through the two board slots next to the holder keeps the cell in during the drop test.
F1 – a 5x20 mm 4 A fast (F4A) glass fuse in two clips: it blows on a short, and it is pulled out and replaced by hand (keep the spare from the BoM in the box). 4 A: the box draws <= 1.8 A at a nearly empty cell.
D1 – normally does nothing (it points 'backwards'). If the cell is put in the holder the wrong way round, D1 conducts and shorts it: F1 blows (or the cell's protection cuts) before any module sees −4 V.
Bench check: D1's leakage at 4.2 V adds to the standby drain (datasheet up to mA at 40 V and hot; measure the cell current with the box off).
C1 – 100 uF reservoir at the boost input (Pololu asks for >= 33 uF).

2 CHARGER LEAD (Adafruit Micro-Lipo USB-C #4410 at the box wall)



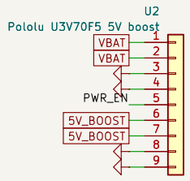
The Adafruit #4410 charger (24-Oct; the TP4056 boards were out of stock) sits at the box wall where its USB-C port is reachable; its BAT and GND pads go to this 2-pin lead. Close its 500 mA solder jumper (default 100 mA): a full 3500 mAh cell then takes ~7-8 h (overnight). Any USB-C cable or charger works.
It has no cell protection of its own: the protected 18650 has it. Charge with the box switched OFF.
J23 is behind F1: a reversed cell never reaches it.

3 POWER TOGGLE (tells the boost to sleep)



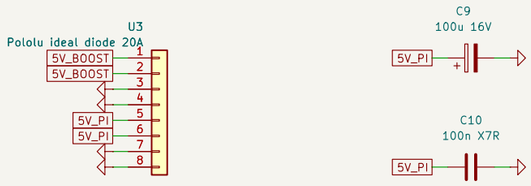
Toggle CLOSED = ENABLE pulled to GND = boost asleep = box OFF (the battery current never goes through the switch). Toggle OPEN = the module's own 100 k pull-up wakes it = box ON. A lead that breaks in the drop test therefore leaves the box running. Mount the toggle so that 'down' (closed) is marked OFF.
C18 + R9: 100 nF at the lead, then 1 k to ENABLE: a zap on the lever goes into C18, R9 limits what reaches the module (C18 sits at the connector since checkpoint 3: room on the board). D16: TVS at the lead (after checkpoint 3), P4KE10A, stand-off 8.55 V > the cell's 4.2 V; its leakage (<= 10 uA even at 8.55 V, far less at 4.2 V) would drop at most 1 V across the 100 k pull-up, so ENABLE still reads high (box ON).
With the toggle closed: EN = VBAT x 1k / 101k = 0.04 V (< 0.4 V needed). Off drain: ~40 uA in the pull-up.

4 5 V BOOST MODULE (Pololu U3V70F5)



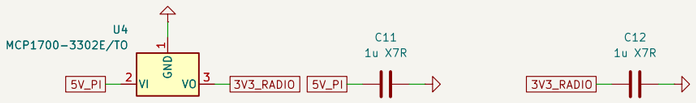
This module turns the cell's 3.0-4.2 V into 5.0 V (+-4 %) for the Pi: about 3 A available at a 3.0 V cell, the Pi needs about 1 A at most. Input protected against reversal: below 2.4 V it stops (UVLO).
ENABLE: 100 k pull-up to VIN on the module: < 0.4 V = true shutdown (its output is disconnected).
Keep the cell -> F1 -> U2 -> GND loop short and wide (1.5-2 mm tracks); keep U2 away from the radio.

5 ONE-WAY VALVE FOR THE PI'S 5 V (Pololu ideal diode #5388)



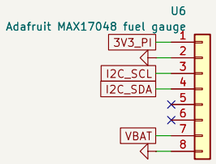
U3 lets current flow from the boost to the Pi but never back (< 3 mOhm, so no voltage is lost). If someone plugs the Pi's micro-USB while the box is on, the USB supply cannot push current into the boost or the cell. It needs ~3.9 V to start, so it only works on the 5 V side – never on the cell.
C9 (100 uF) + C10 (100 nF) sit at the box header's pins 2/4: the Pi's supply reservoir.
Rule all the same: do not power the Pi from its micro-USB AND the box at once (two supplies share the Pi).

6 RADIO 3.3 V: MCP1700-3302E/TO from the Pi's 5 V



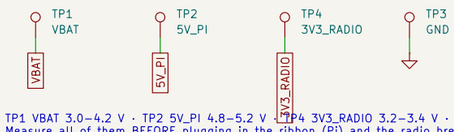
A 3-legged regulator: 5 V in, always exactly 3.3 V out for the radio (max 250 mA; the radio takes <= 120 mA). Fed from 5V_PI, so the radio is only powered when the Pi is: it can never push current into a switched-off Pi through its pins. 0.2 W at full transmit: warm, not hot. Never feed the radio from the cell or the Pi's 3.3 V pin (absolute max 3.6 V). C11/C12: 1 uF ceramics (datasheet: >= 1 uF at the output); the breakout adds 100 uF + 10 uF + 100 nF at the module. Bench: ripple on 3V3_RADIO during transmit (scope, TP4).

7 BATTERY LEVEL (Adafruit MAX17048 #5580, I2C 0x36)

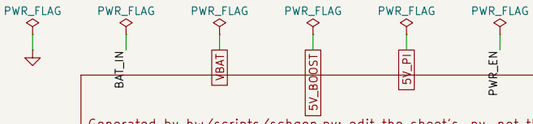


Runs from the cell: its chip (VDD) must sit on BAT through the 3-pad jumper SJ2 (centre pad = VDD, one side VIN, the other BAT). Adafruit says it runs with 3.3 or 5 V on VIN, so it ships VDD-BAT, but the GitHub board file shows SJ2 open (review 24-Oct): ON ARRIVAL check with a meter that SJ2's centre pad connects to BAT; if it does not, bridge centre-to-BAT with solder. VIN = the Pi's 3.3 V (3V3_PI, header pin 1): only its 10 k I2C/INT pull-ups and its LED. BAT / BAT_GND = the JP2 pair, straight at VBAT. INT, QSTART unused: the software polls the gauge over I2C (0x36). Never set MODE.EnSleep in software.

8 TEST POINTS (clip a multimeter here)



TP1 VBAT 3.0-4.2 V · TP2 5V_PI 4.8-5.2 V · TP4 3V3_RADIO 3.2-3.4 V · TP3 GND (black).
Measure all of them BEFORE plugging in the ribbon (Pi) and the radio breakout. TP2/TP4 are low wire loops (~5 mm) under the breakout's place (a Keystone 5010 is too tall there); TP3 (GND) is a loop too, beside TP1.



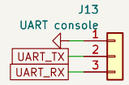
Generated by hw/scripts/schgen.py; edit the sheet's .py, not this file
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Sheet: /power/		
File: power.kicad_sch		
Title: Power: cell, fuse, 5 V boost, ideal diode, radio 3.3 V, fuel gauge (rev		
Size: A3	Date: 2026-10-24	**Rev: DRAFT**
KiCad E.D.A. 10.0.6		Id: 2/5

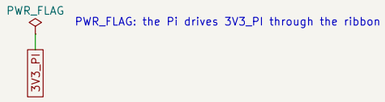
PI LINK - Raspberry Pi 3 Model A+ on a 40-wire ribbon

J1 is a shrouded (keyed) 2x20 box header; a 15 cm 40-wire female-female ribbon goes to the Pi's GPIO header.
The key stops it going in rotated. Red wire = pin 1 at both ends.
5V_PI (pins 2, 4) comes from the ideal diode on the power sheet. All eight GND pins go to the carrier ground:
they are the return path of the Pi's supply and of the SPI signals.
3V3_PI = the Pi's own 3.3 V, from pin 1 only (pin 17 unused): it only feeds the fuel gauge's I2C pull-ups.
Pins 27/28 (ID EEPROM) stay unconnected. The Pi's WiFi/BT chip is disabled or removed (Atenea Q22).
Do not power the Pi from its micro-USB while the box is on (the ideal diode protects the boost, but two
supplies would share the Pi). For development: unplug the ribbon, or switch the box off.

UART CONSOLE (bench)



1 GND, 2 TX (Pi GPIO 14 out), 3 RX (GPIO 15 in). 3.3 V USB-serial adapters only (never 5 V).



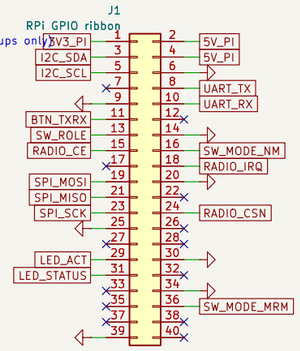
PWR_FLAG: the Pi drives 3V3_PI through the ribbon

MOUNTING (M3, unplated, no copper around)



Header pin / BCM function (pinout.xyz)

1	3V3 (feeds 3V3_PI: the fuel gauge's I2C pull-ups only)
3	GPIO 2 SDA1
5	GPIO 3 SCL1
7	GPIO 4
9	GND
11	GPIO 17 GO/STOP
13	GPIO 27 role
15	GPIO 22 radio CE
17	3V3 (unused: pin 1 is used)
19	GPIO 10 MOSI0
21	GPIO 9 MISO0
23	GPIO 11 SCLK0
25	GND
27	ID_SD: keep NC
29	GPIO 5 LED activity
31	GPIO 6 LED status
33	GPIO 13
35	GPIO 19
37	GPIO 26 (not used)
39	GND



Header pin / BCM function

2	5V
4	5V
6	GND
8	GPIO 14 TXD0
10	GPIO 15 RXD0
12	GPIO 18
14	GND
16	GPIO 23 mode NM
18	GPIO 24 radio IRQ
20	GND
22	GPIO 25 (FG_ALRT, not used)
24	GPIO 8 CE0 = radio CSN
26	GPIO 7 CE1
28	ID_SC: keep NC
30	GND
32	GPIO 12
34	GND
36	GPIO 16 mode MRM
38	GPIO 20
40	GPIO 21

WIRE LINKS (one copper layer): INSULATED 22 AWG solid tinned copper wire on the component side between two unplated 1.0 mm holes, ends stripped ~3 mm, soldered on the copper side; lay it flat, glue it for the drop test. Positions and lengths: hw/teamc-carrier-simple/links.json and the assembly drawing.



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Sheet: /pi_link/
File: pi_link.kicad_sch

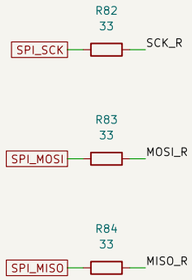
Title: Pi link: 2x20 box header + ribbon, UART, holes (rev C, DRAFT)

Size: A3	Date: 2026-10-24	**Rev: DRAFT**
KiCad E.D.A. 10.0.6		Id: 3/5

RADIO - socket for the lab-made E01-ML01SP4 breakout

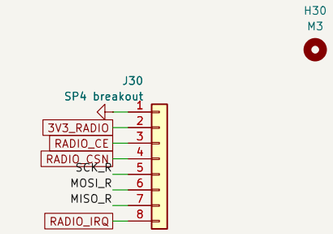
The breakout plugs into J30 (its pins down, the SP4 on top) and is held by an M3 standoff at H30.
VCC = 3V3_RADIO from the MCP1700 on the power sheet only (radio range 2.0-3.6 V, absolute max 3.6 V).
Put J30 at the board end nearest the antenna wall and away from the boost module; the flex antenna's
U.FL plug goes onto the SP4 (glue the plug for the drop test).

SPI SERIES RESISTORS (33 Ω, through-hole)



33 Ω in SCK, MOSI and MISO soften the fast edges that travel down the 15 cm ribbon (less ringing).
On this one-layer board their legs are 7.6 mm apart, so another track may pass under a resistor body
between its legs: the board drawing says which. A 0 Ω wire link can replace one if the bench shows no need.

BREAKOUT SOCKET



CSN PULL-UP



R80 keeps CSN high (radio not selected) while the Pi boots. 3V3_RADIO and the Pi's 3.3 V come up and go
down together (both from 5V_PI). No IRQ pull-up: the Linux HAL turns on the Pi's own pull-up on GPIO 24
(IRQ is open drain, falling edge).

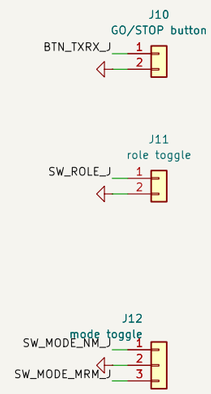
Generated by hw/scripts/schgen.py; edit the sheet's .py, not this file
DRAFT - for ET review
MTP F26 Team C (UPC)

Sheet: /radio/		
File: radio.kicad_sch		
Title: Radio: SP4 breakout socket (rev C, DRAFT)		
Size: A3	Date: 2026-10-24	**Rev: DRAFT**
KiCad E.D.A. 10.0.6		Id: 4/5

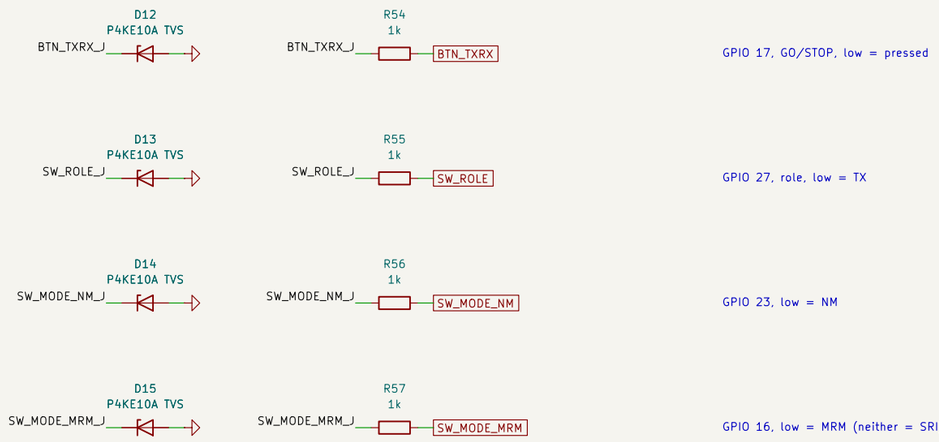
UI - panel controls and LEDs on JST-XH leads

Every panel part sits on the box wall and plugs into the board with a JST-XH lead (glue the housings and tie the leads down for the drop test). Inputs: the switch pulls the GPIO to GND; the Pi's own pull-up (turned on by ui.c) holds it high otherwise. A TVS at each lead + 1 kΩ in series protect the GPIO from a zap and from a GPIO driven by old firmware at boot (GPIO 16). LEDs light when the GPIO is high.
Panel holes: Ø7.5 mm for the PS1024 button, Ø6.5 mm for the 100-series toggles, Ø8 mm for the LED clips.

CONNECTORS (panel parts off-board)



TVS AT THE LEAD + SERIES 1 kΩ

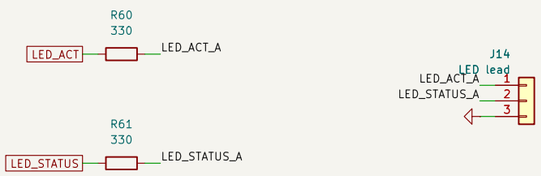


J10: GO/STOP button (also the bench button: bridge its two pins).
J11: role toggle, closed = TX, open = RX (may stay unfitted until Atenea Q8).
J12: mode toggle ON-OFF-ON: 1 NM, 2 common (GND), 3 MRM; centre = SRI.

D12–D15 (Santiago, checkpoint 3): one TVS per panel line, right at its lead connector, cathode (band) to the line, anode to GND. A zap on a metal lever goes into the TVS (it clamps at ~14,5 V), and the 1 kΩ then limits what reaches the Pi's own protection diodes to ~10 mA for a few nanoseconds. P4KE10A: stand-off 8.55 V, leakage at most 10 µA (at 8.55 V; far less at 3.3 V), so the Pi's ~50 kΩ pull-up still reads high. Stands upright.

Layout change at checkpoint 3: the 100 nF capacitors at the GPIOs (C50–C53 of checkpoint 2) are gone. On one copper layer each needed a ground return in a crowded place (and two of them a wire link). The switches are slow and the software debounces them (30 ms, ui.c); the 1 kΩ still limits the current of a zap or of a GPIO driven at boot to a few mA into the Pi's own protection diodes.

LEDs (5 mm, in panel clips; ≈ 4 mA from 3.3 V)



J14 lead: pin 1 to the green LED's long leg (activity, GPIO 5), pin 2 to the red LED's long leg (status, GPIO 6), pin 3 to both short legs. (3.3 V − ~2.0 V) / 330 Ω ≈ 4 mA, far under the Pi's 16 mA per pin. Red/green/yellow only (blue/white need more voltage than 3.3 V leaves).

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Sheet: /ui/		
File: ui.kicad_sch		
Title: UI: panel button, toggles, LEDs (rev C, DRAFT)		
Size: A3	Date: 2026-10-24	**Rev: DRAFT**
KiCad E.D.A. 10.0.6		Id: 5/5